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|---|---|---|
| Marwadi University | **Marwadi University**<br>**Faculty of Technology**<br>**Department of Information and Communication Technology** | |
| **Subject: Digital Signal and Image Processing (01CT1513)** | **Aim : Design Butterworth and Chebyshev filter using bilinear transformation method.** | |
| **Experiment : 4** | **Date: 10/10/2026** | **Enrolment No:92400133057** |

Github :- https://github.com/manavkasundra/DSIP

Code :-

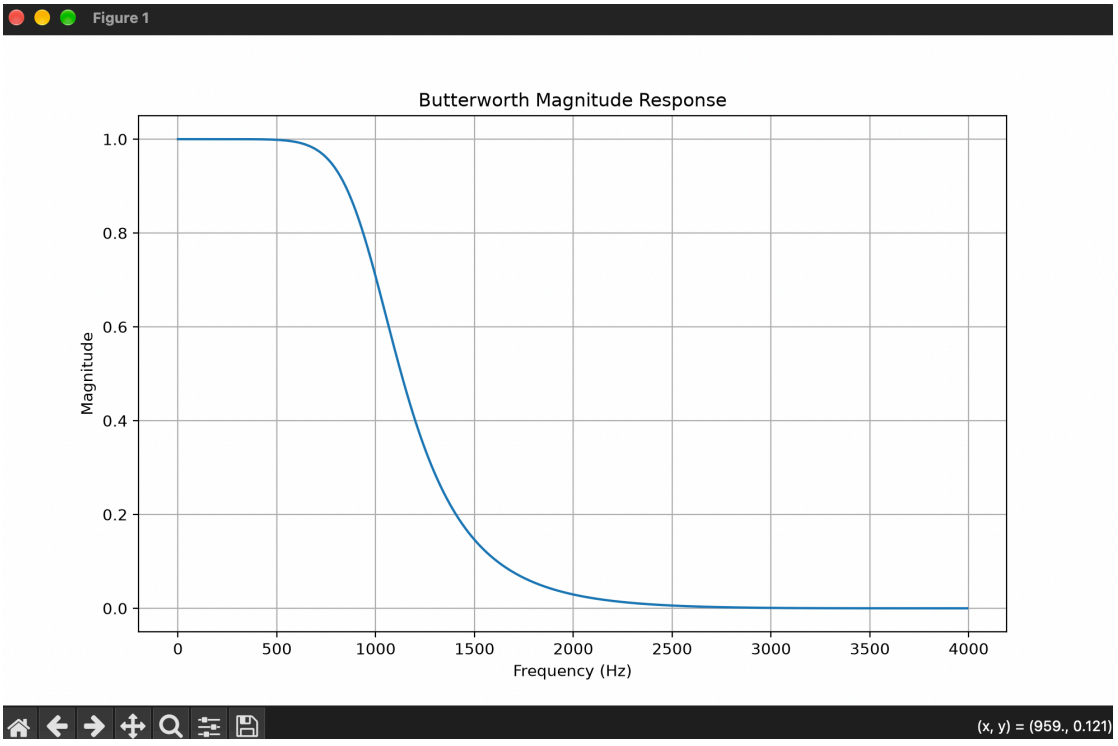
```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import butter, cheby1, freqz, lfilter
def design_butterworth_filter(filter_order, cutoff_frequency, sampling_frequency):
    return butter(
        filter_order,
        cutoff_frequency,
        btype="low",
        fs=sampling_frequency,
    )
def design_chebyshev_filter(
    filter_order, cutoff_frequency, sampling_frequency, ripple
):
    return cheby1(
        filter_order,
        ripple,
        cutoff_frequency,
        btype="low",
        fs=sampling_frequency,
    )
def plot_filter_response(digital_b, digital_a, sampling_frequency, title):
    frequency, response = freqz(digital_b, digital_a, fs=sampling_frequency)
    plt.figure(figsize=(10, 6))
    plt.plot(frequency, np.abs(response))
    plt.title(f"{title} Magnitude Response")
    plt.xlabel("Frequency (Hz)")
    plt.ylabel("Magnitude")
    plt.grid(True)
    plt.show()
    impulse = np.zeros(4096)
    impulse[0] = 1
    impulse_response = lfilter(digital_b, digital_a, impulse)
    plt.figure(figsize=(10, 6))
    plt.plot(impulse_response)
    plt.title(f"{title} Impulse Response")
    plt.xlabel("Samples")
    plt.ylabel("Amplitude")
    plt.grid(True)
    plt.show()
filter_order = 4
cutoff_frequency = 1000
```

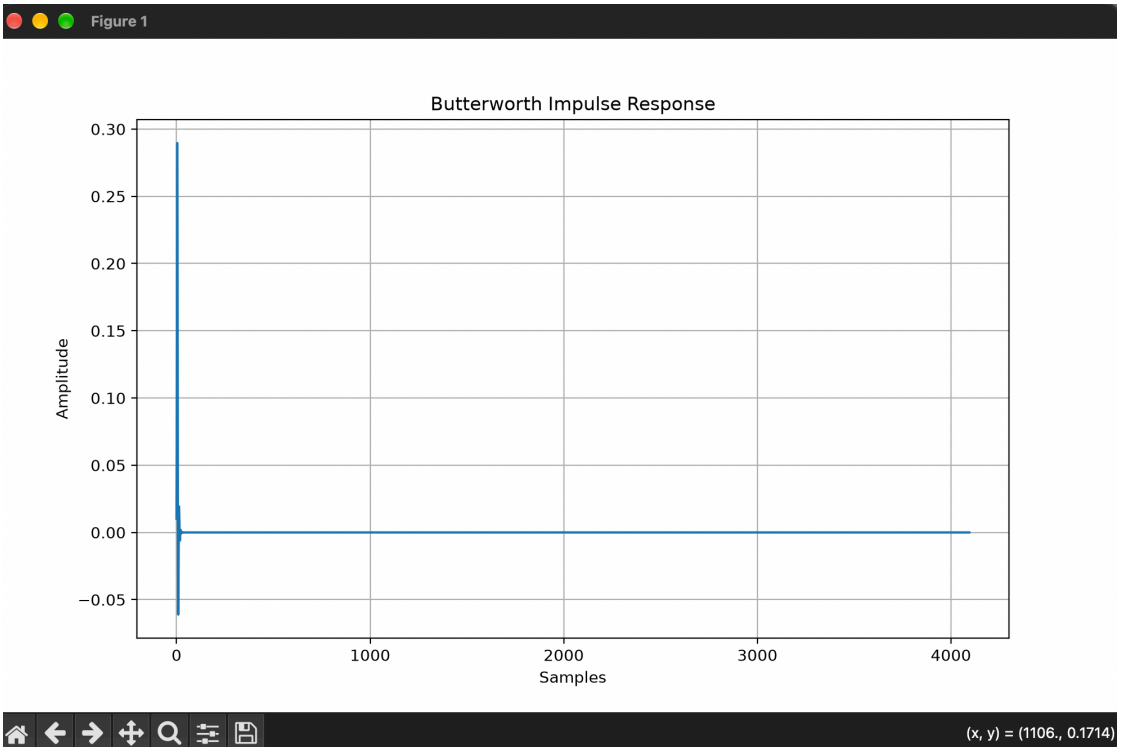
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```
sampling_frequency = 8000
ripple = 0.5
butter_b, butter_a = design_butterworth_filter(
   filter_order, cutoff_frequency, sampling_frequency
)
plot_filter_response(butter_b, butter_a, sampling_frequency, "Butterworth")
cheby_b, cheby_a = design_chebyshev_filter(
   filter_order, cutoff_frequency, sampling_frequency, ripple
)
plot_filter_response(cheby_b, cheby_a, sampling_frequency, "Chebyshev")
filter_path = "filter_coefficients.txt"
np.savetxt(filter_path, np.vstack((cheby_b, cheby_a)), delimiter=",")
print(f"Filter coefficients saved at: {filter_path}")
```

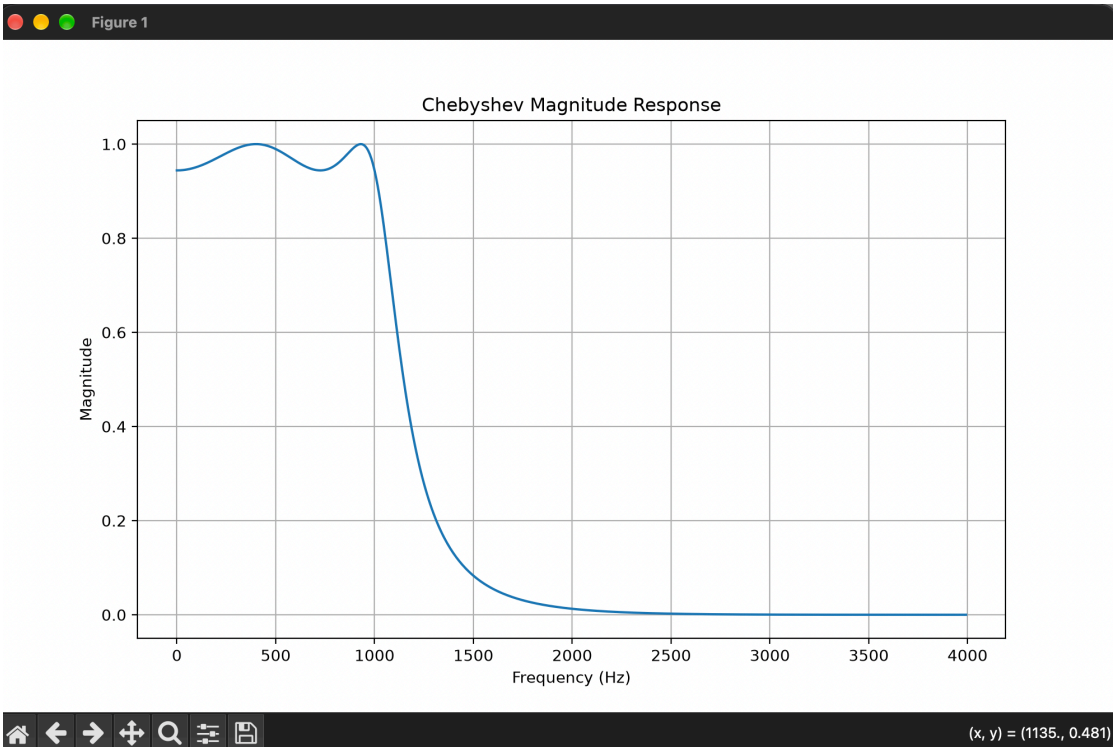
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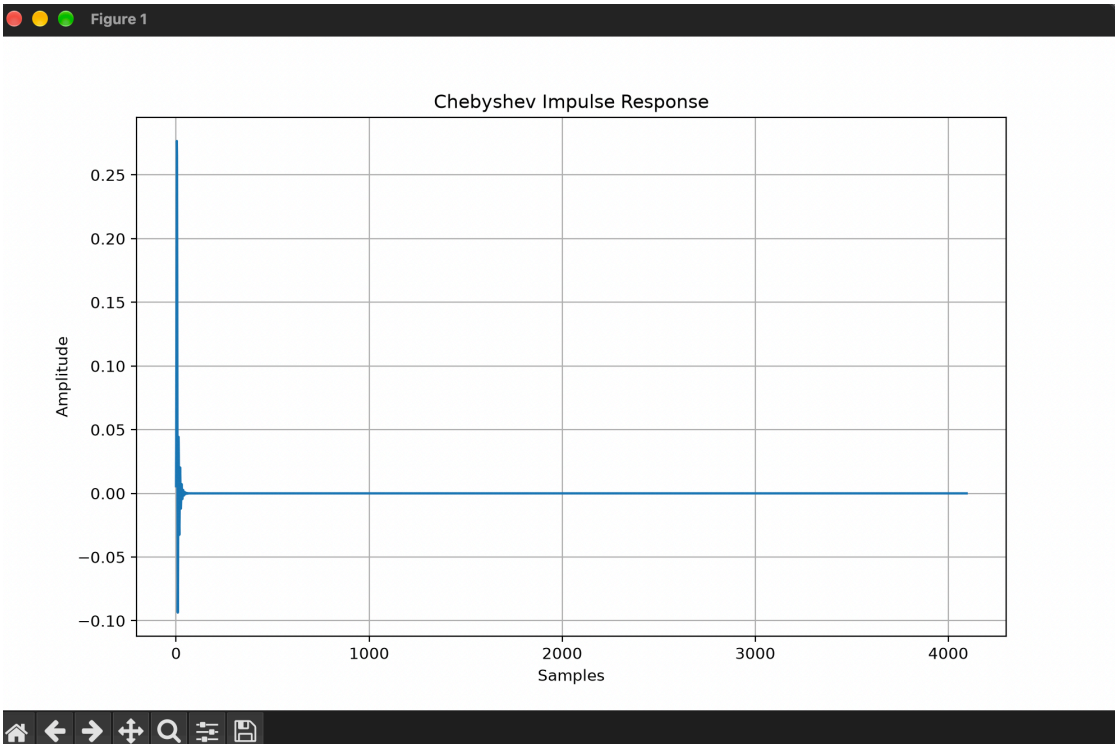
Output :-



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# Additional Task

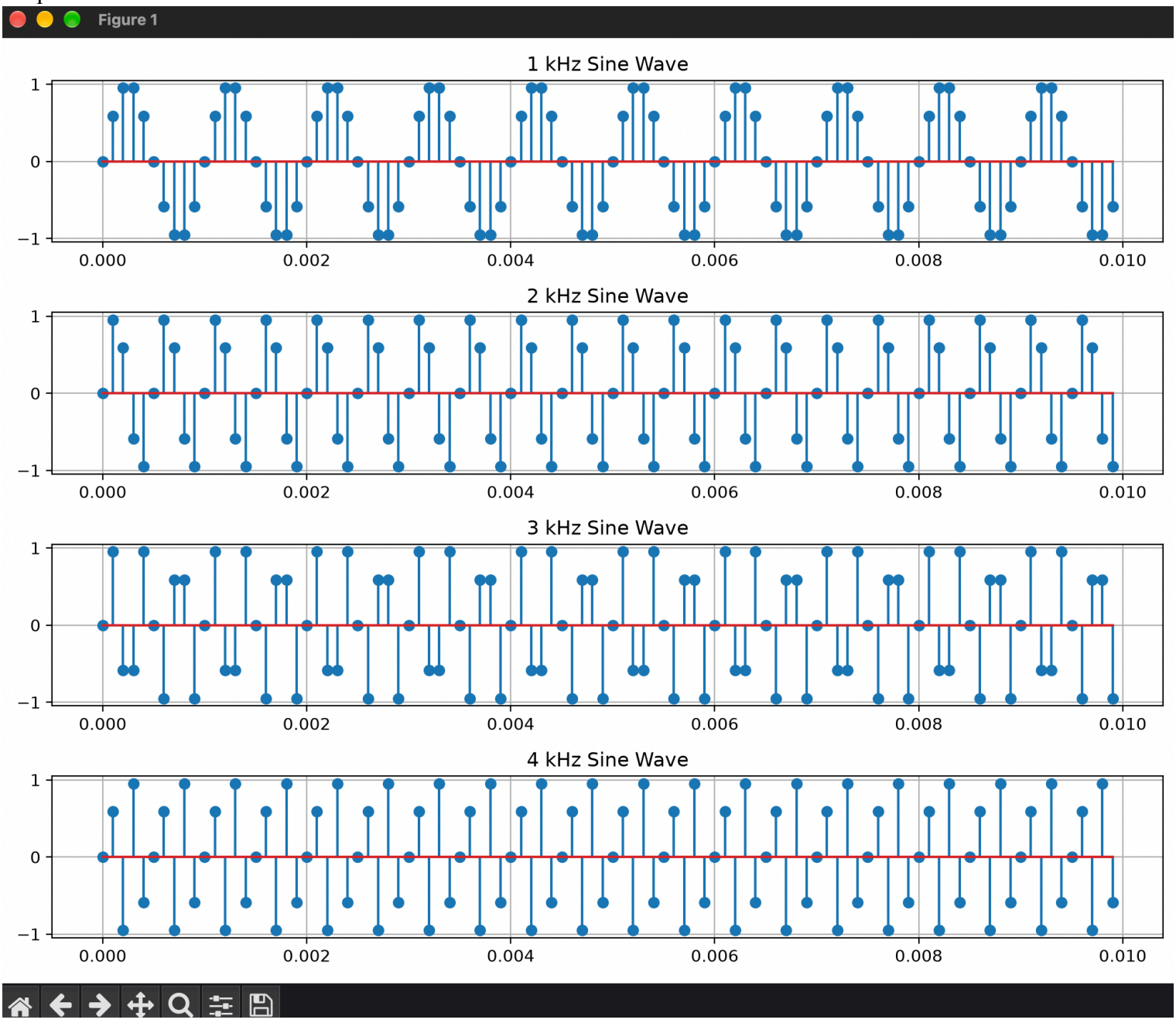
•Generate a discrete sine wave of 1KHz, 2KHz, 3KHz, 4 KHz. Plot all these Code :

Code:-

```
import numpy as np
import matplotlib.pyplot as plt
fs = 10000
t = np.arange(0, 0.01, 1/fs)
x1 = np.sin(2*np.pi*1000*t)
x2 = np.sin(2*np.pi*2000*t)
x3 = np.sin(2*np.pi*3000*t)
x4 = np.sin(2*np.pi*4000*t)
plt.figure(figsize=(10,8))
plt.subplot(4,1,1)
plt.stem(t,x1)
plt.title("1 kHz Sine Wave")
plt.grid()
plt.subplot(4,1,2)
plt.stem(t,x2)
plt.title("2 kHz Sine Wave")
plt.grid()
plt.subplot(4,1,3)
plt.stem(t,x3)
plt.title("3 kHz Sine Wave")
plt.grid()
plt.subplot(4,1,4)
plt.stem(t,x4)
plt.title("4 kHz Sine Wave")
plt.grid()
plt.tight_layout()
plt.show()
```

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Output:-



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•Add all these four signal and plot it in time as well as frequency domain

Code :-

```
import numpy as np

import matplotlib.pyplot as plt

fs = 10000

t = np.arange(0, 0.01, 1 / fs)

x1 = np.sin(2 * np.pi * 1000 * t)

x2 = np.sin(2 * np.pi * 2000 * t)

x3 = np.sin(2 * np.pi * 3000 * t)

x4 = np.sin(2 * np.pi * 4000 * t)

x = x1 + x2 + x3 + x4

# Time domain

plt.figure(figsize=(10, 4))

plt.stem(t, x)

plt.title("Added Signal - Time Domain")

plt.xlabel("Time (s)")

plt.ylabel("Amplitude")

plt.grid(True)

plt.show()

# Frequency domain

X = np.fft.fft(x)

freq = np.fft.fftfreq(len(x), 1 / fs)

plt.figure(figsize=(10, 4))

positive_freqs = freq[: len(freq) // 2]

positive_magnitudes = np.abs(X[: len(X) // 2])

plt.stem(positive_freqs, positive_magnitudes)

plt.title("Added Signal - Frequency Domain")

plt.xlabel("Frequency (Hz)")

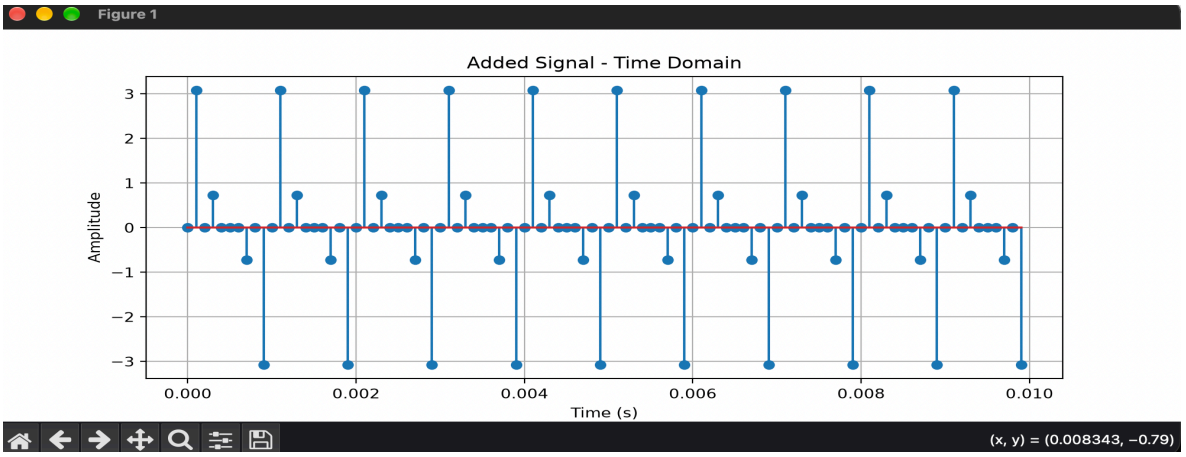
plt.ylabel("Magnitude")

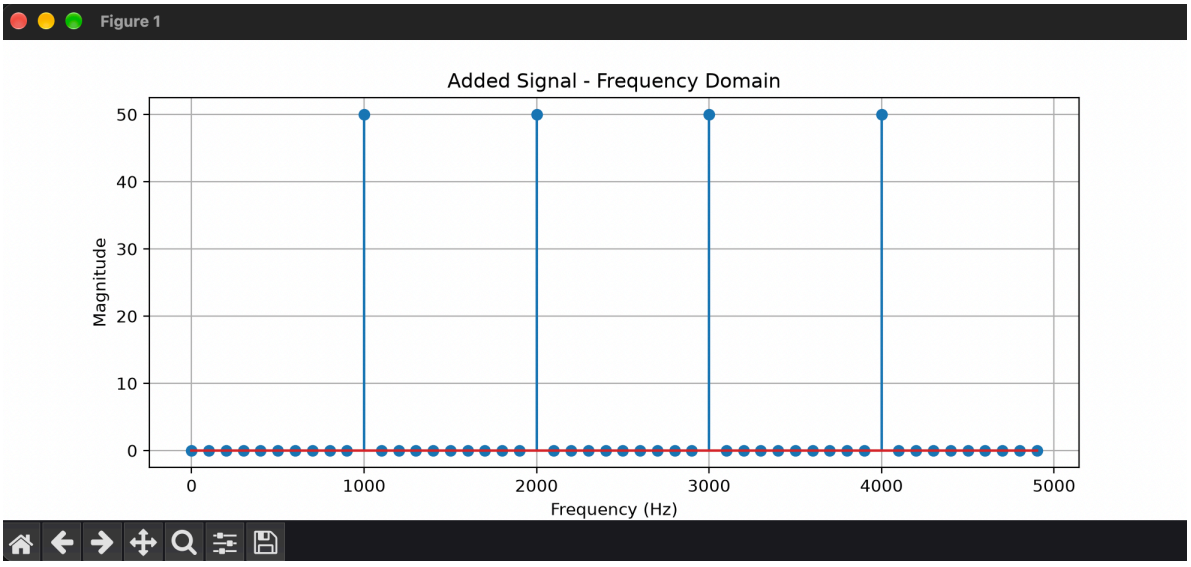
plt.grid(True)

plt.show()
```

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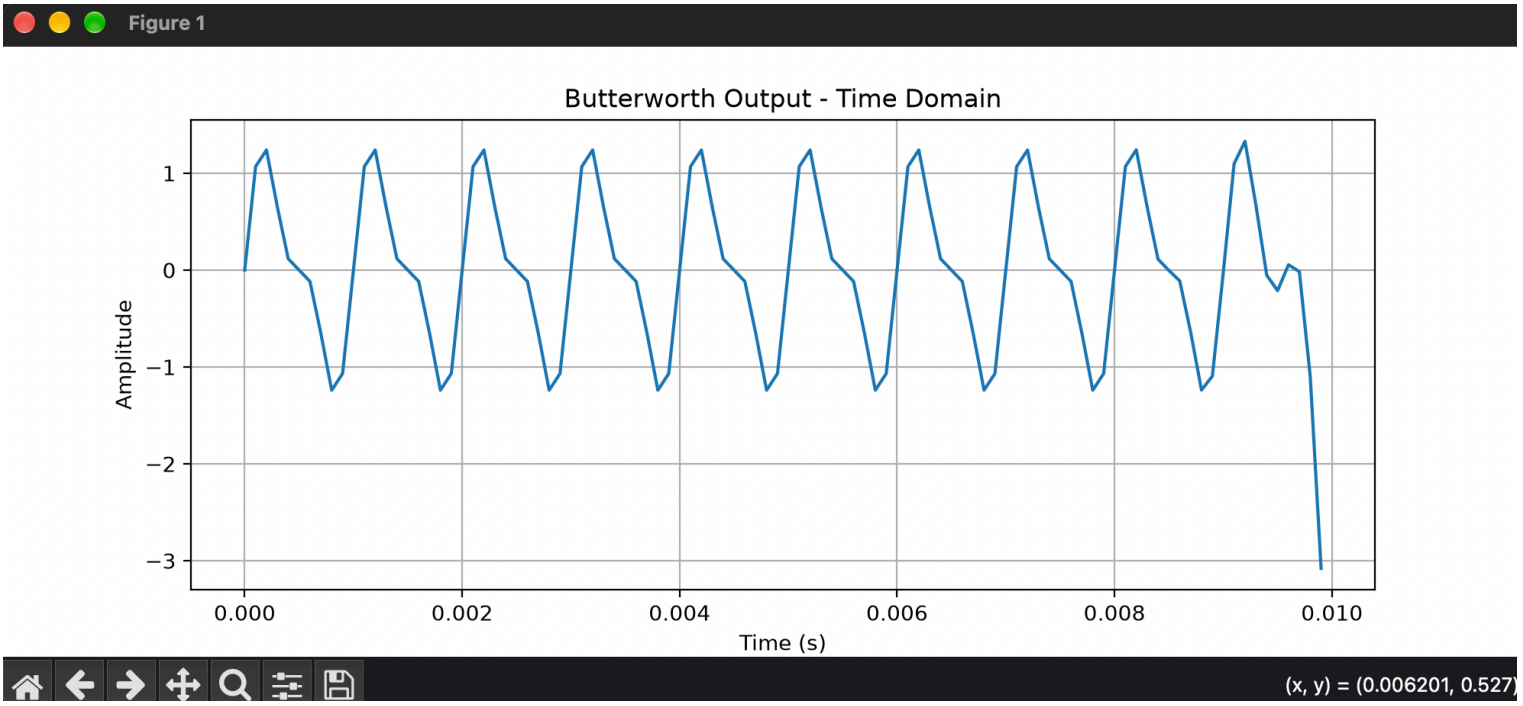
pass through buttorworth digital filter having cutoff frequency 2 khz and plot the output plot in time as well as frequency domain

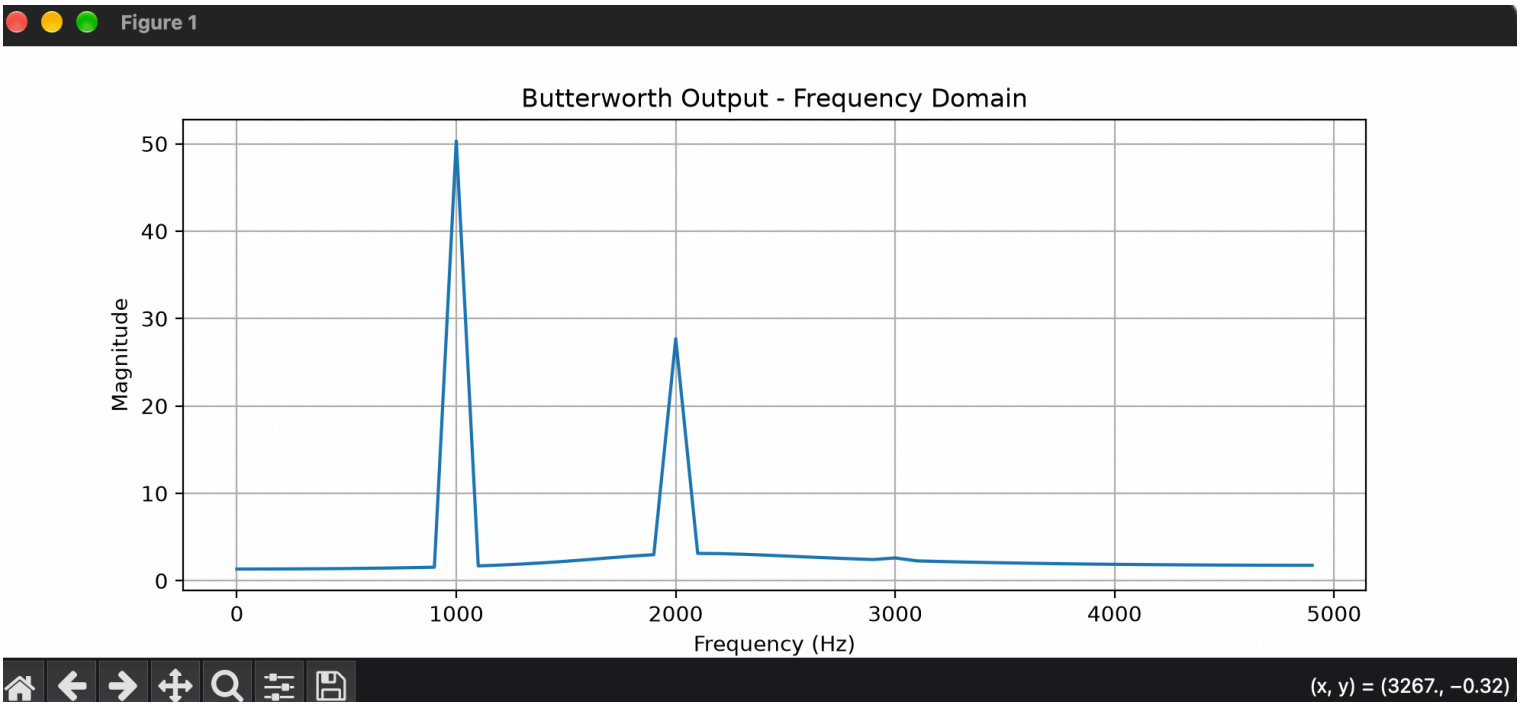
Code :-

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import butter, filtfilt
fs = 10000
t = np.arange(0, 0.01, 1 / fs)
x1 = np.sin(2 * np.pi * 1000 * t)
x2 = np.sin(2 * np.pi * 2000 * t)
x3 = np.sin(2 * np.pi * 3000 * t)
x4 = np.sin(2 * np.pi * 4000 * t)
x = x1 + x2 + x3 + x4
# Apply a fourth-order Butterworth low-pass filter
b, a = butter(4, 2000 / (fs / 2), btype="low")
y = filtfilt(b, a, x)
# Time domain
plt.figure(figsize=(10, 4))
plt.plot(t, y)
plt.title("Butterworth Output - Time Domain")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.grid(True)
plt.show()
# Frequency domain
Y = np.fft.fft(y)
freq = np.fft.fftfreq(len(y), 1 / fs)
plt.figure(figsize=(10, 4))
plt.plot(freq[: len(freq) // 2], np.abs(Y[: len(Y) // 2]))
plt.title("Butterworth Output - Frequency Domain")
plt.xlabel("Frequency (Hz)")
plt.ylabel("Magnitude")
plt.grid(True)
plt.show()
```

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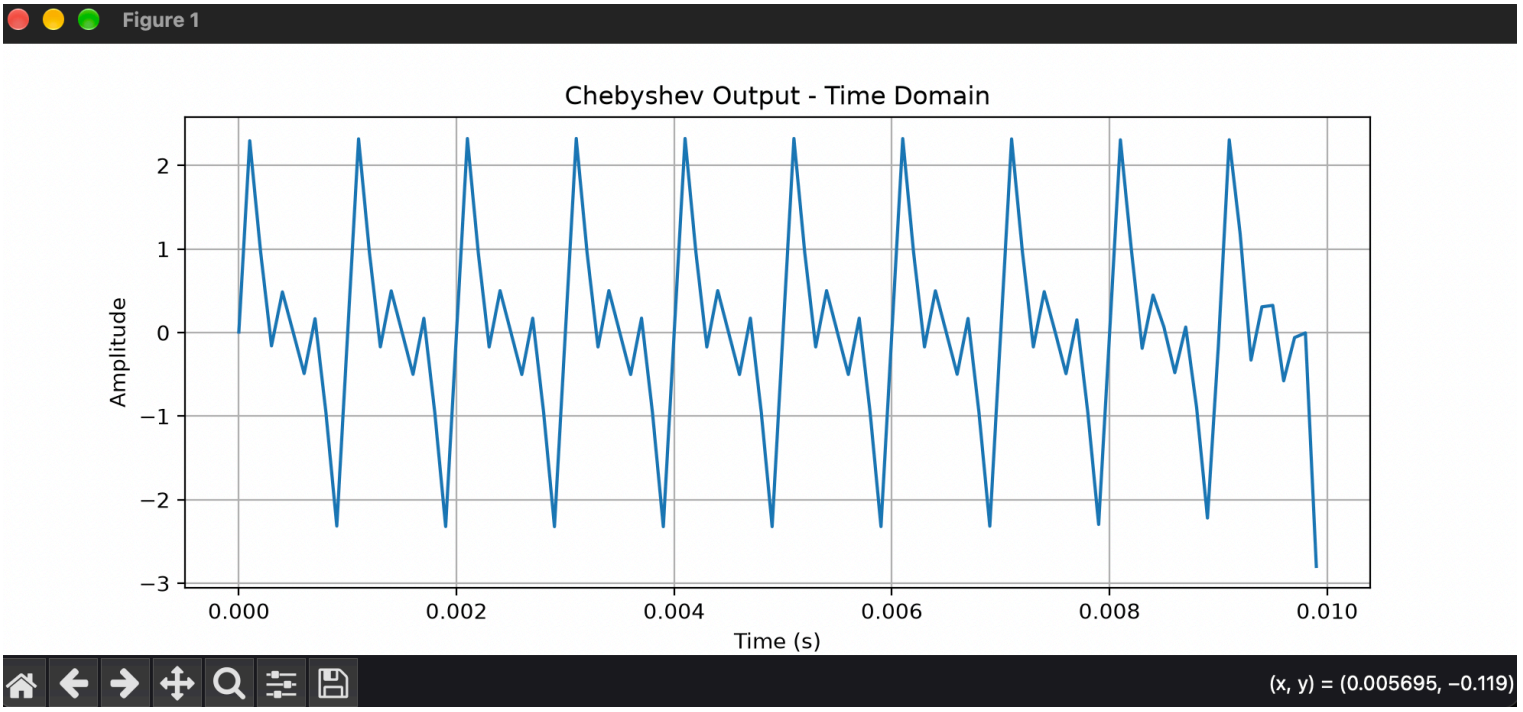
•Pass through chebyschev digital filter having cutoff frequency 3KHz and plot the output plot in time as well as frequency domain

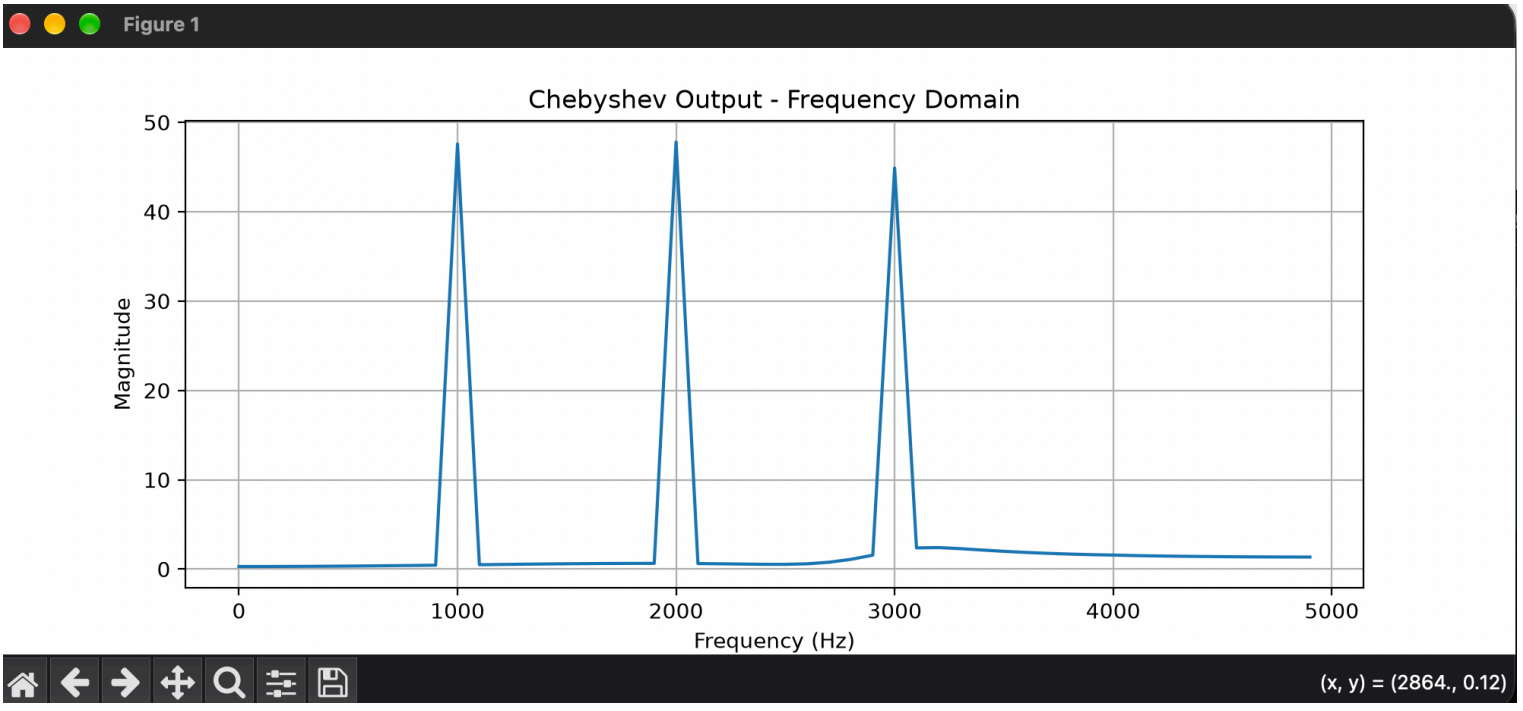
Code :-

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import cheby1, filtfilt
fs = 10000
t = np.arange(0, 0.01, 1 / fs)
x1 = np.sin(2 * np.pi * 1000 * t)
x2 = np.sin(2 * np.pi * 2000 * t)
x3 = np.sin(2 * np.pi * 3000 * t)
x4 = np.sin(2 * np.pi * 4000 * t)
x = x1 + x2 + x3 + x4
# Apply a fourth-order Chebyshev Type I low-pass filter
b, a = cheby1(4, 0.5, 3000 / (fs / 2), btype="low")
y = filtfilt(b, a, x)
# Time domain
plt.figure(figsize=(10, 4))
plt.plot(t, y)
plt.title("Chebyshev Output - Time Domain")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.grid(True)
plt.show()
# Frequency domain
Y = np.fft.fft(y)
freq = np.fft.fftfreq(len(y), 1 / fs)
plt.figure(figsize=(10, 4))
plt.plot(freq[: len(freq) // 2], np.abs(Y[: len(Y) // 2]))
plt.title("Chebyshev Output - Frequency Domain")
plt.xlabel("Frequency (Hz)")
plt.ylabel("Magnitude")
plt.grid(True)
plt.show()
```

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Conclusion :-

In this experiment, we designed and evaluated Butterworth and Chebyshev Type I digital low-pass filters, examining their magnitude and impulse responses to get a clear picture of their frequency characteristics and performance. To test them in a realistic scenario, we created a combined test signal by adding together four individual sine waves at 1 kHz, 2 kHz, 3 kHz, and 4 kHz, analyzing the result in both the time and frequency domains using the Fast Fourier Transform (FFT).

Running the composite signal through both filters highlighted the practical differences in how they process frequencies:

- The Butterworth low-pass filter (set to a 2 kHz cutoff) allowed the 1 kHz and 2 kHz frequency components to pass through smoothly while cleanly filtering out the higher 3 kHz and 4 kHz components.

The Chebyshev Type I low-pass filter (configured with a 3 kHz cutoff) preserved the 1 kHz, 2 kHz, and 3 kHz components, selectively attenuating only the 4 kHz component as intended.